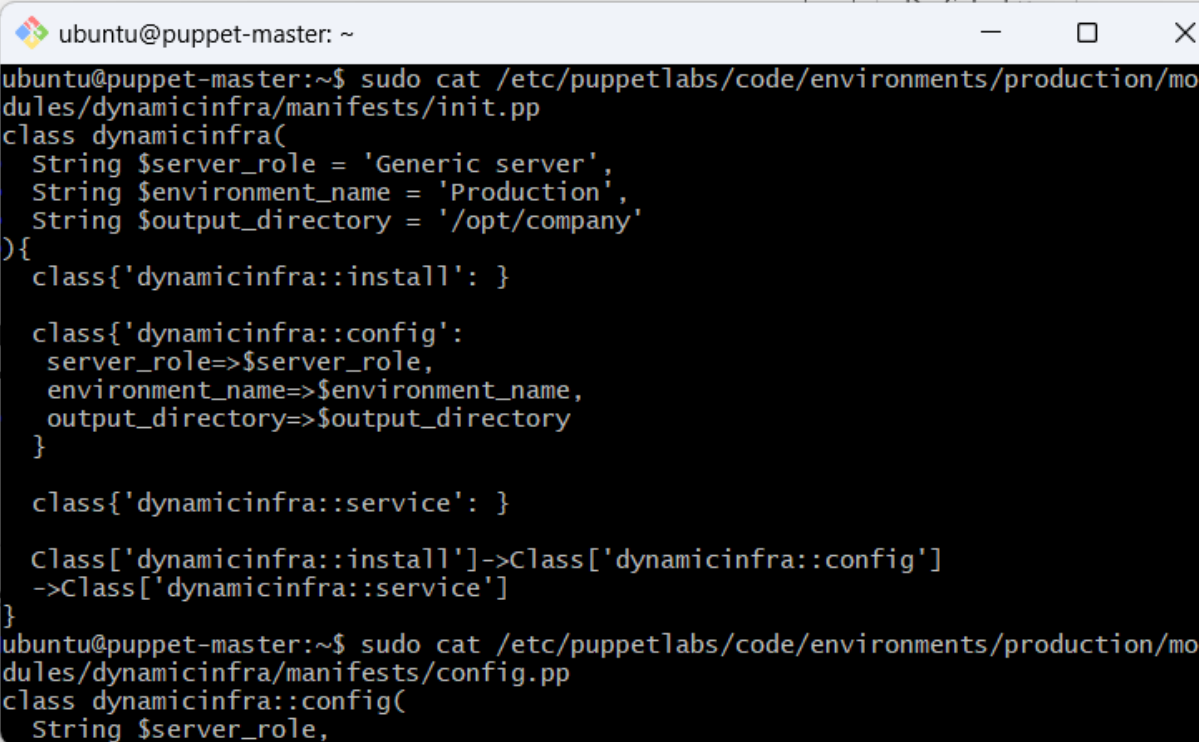


Complete Puppet Mastery Part-3(Flagship Standalone Practical)

Practical 12 - Enterprise Dynamic Configuration Management Using Puppet Facts, EPP Templates, Parameterized Classes, and Centralized Modules

Master:

A terminal window titled 'ubuntu@puppet-master: ~' with standard window controls. It shows the execution of two Puppet manifest files. The first command is 'sudo cat /etc/puppetlabs/code/environments/production/modules/dynamicinfra/manifests/init.pp', which displays a Puppet class definition for 'dynamicinfra'. This class has three parameters: 'server_role' (default 'Generic server'), 'environment_name' (default 'Production'), and 'output_directory' (default '/opt/company'). It defines three subclasses: 'dynamicinfra::install', 'dynamicinfra::config', and 'dynamicinfra::service'. The 'install' subclass includes a reference to the 'config' subclass. The second command is 'sudo cat /etc/puppetlabs/code/environments/production/modules/dynamicinfra/manifests/config.pp', which shows the beginning of a subclass definition for 'dynamicinfra::config' with a parameter 'server_role'.

```
ubuntu@puppet-master: ~  
String $environment_name,  
String $output_directory  
{  
$service_name = $facts['os']['family'] ? {  
  'Debian'=>'apache2',  
  'RedHat'=>'httpd',  
  default=>'apache2'  
}  
  
file{$output_directory:  
  ensure=>directory,  
  owner=>'root',  
  group=>'root',  
  mode=>'0755'  
}  
  
file{"${output_directory}/system-info.txt":  
  ensure=>file,  
  owner=>'root',  
  group=>'root',  
  mode=>'0644',  
  
  content=>epp(  
    content=>epp(  
      'dynamicinfra/system-info.epp',  
      {  
        'server_role'=>$server_role,  
        'environment_name'=>$environment_name  
      }  
    ),  
    notify=>Service[$service_name]  
  }  
  
  if $facts['os']['family'] == 'Debian' {  
    file{"${output_directory}/os-info.txt":  
      ensure=>file,  
      owner=>'root',  
      group=>'root',  
      mode=>'0644',  
      content=>"This node belongs to the Debian family.  
Package Manager: APT\n"  
    }  
  }  
  elseif $facts['os']['family'] == 'RedHat' {  
    file{"${output_directory}/os-info.txt":
```

```
ubuntu@puppet-master: ~  
content=>epp(  
  'dynamicinfra/system-info.epp',  
  {  
    'server_role'=>$server_role,  
    'environment_name'=>$environment_name  
  }  
)  
),  
notify=>Service[$service_name]  
}  
  
if $facts['os']['family'] == 'Debian' {  
  file{"${output_directory}/os-info.txt":  
    ensure=>file,  
    owner=>'root',  
    group=>'root',  
    mode=>'0644',  
    content=>"This node belongs to the Debian family.  
Package Manager: APT\n"  
  }  
}  
elseif $facts['os']['family'] == 'RedHat' {  
  file{"${output_directory}/os-info.txt":
```

```
ubuntu@puppet-master: ~  
    ensure=>file,  
    owner=>'root',  
    group=>'root',  
    mode=>'0644',  
    content=>"This node belongs to the Redhat family.\nPackage Manager:  
    DNF/YUM\n"  
  }  
}  
}  
ubuntu@puppet-master:~$ sudo cat /etc/puppetlabs/code/environments/production/modules/dynamicinfra/manifests/service.pp  
class dynamicinfra::service{  
  
  if $facts['os']['family'] == 'Debian' {  
  
    service{'apache2':  
      ensure=>'running',  
      enable=>true  
    }  
  }  
  
  elsif $facts['os']['family'] == 'RedHat'{  
  
    service{'httpd':
```

```
    service{'httpd':  
      ensure=>'running',  
      enable=>true  
    }  
  }  
}  
ubuntu@puppet-master:~$ sudo cat /etc/puppetlabs/code/environments/production/modules/dynamicinfra/manifests/install.pp  
class dynamicinfra::install{  
  
  package{'tree':  
    ensure=>installed  
  }  
  
  package{'curl':  
    ensure=>installed  
  }  
  
  if $facts['os']['family'] == 'Debian' {  
  
    package{'apache2':  
      ensure=>installed  
    }  
  }  
}
```

```
ubuntu@puppet-master: ~  
    elsif $facts['os']['family'] == 'RedHat'{  
        package{'httpd':  
            ensure=>installed  
        }  
    }  
}  
ubuntu@puppet-master:~$ sudo cat /etc/puppetlabs/code/environments/production/manifests/site.pp  
node 'puppet-agent01'{  
    class{'dynamicinfra':  
        server_role=>'Web Server',  
        environment_name=>'Production',  
        output_directory=>'/opt/company'  
    }  
}  
  
node 'puppet-agent02'{  
    class{'dynamicinfra':  
        server_role=>'Application server',  
        environment_name=>'Development',  
        output_directory=>'/opt/company'  
    }  
}  
ubuntu@puppet-master:~$
```

Agent_01:

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1785845417'  
Notice: Applied catalog in 0.45 seconds  
ubuntu@puppet-agent01:~$ ls -l /opt/company  
total 8  
-rw-r--r-- 1 root root 68 Aug  4 12:09 os-info.txt  
-rw-r--r-- 1 root root 543 Aug  4 12:09 system-info.txt  
ubuntu@puppet-agent01:~$ cat /opt/company/system-info.txt  
-----  
Enterprise Infrastructure Report  
-----  
  
Server Role: Web Server  
Environment: Production  
  
Hostname: puppet-agent01  
FQDN: puppet-agent01.ap-south-1.compute.internal  
Primary IP: 172.31.7.84
```

```
ubuntu@puppet-agent01: ~  
-----  
Enterprise Infrastructure Report  
-----  
Server Role: Web Server  
Environment: Production  
  
Hostname: puppet-agent01  
FQDN: puppet-agent01.ap-south-1.compute.internal  
Primary IP: 172.31.7.84  
  
Operating System: Ubuntu  
OS Release: 22.04  
OS Family: Debian  
  
Kernel: Linux  
Architecture:  
  
Processors: 2  
Memory: 914.01 MiB  
  
Puppet Version: 8.10.0  
  
Generated By: Puppet Master and Vishnu's Passion
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1785845938'  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/con  
tent:  
--- /opt/company/system-info.txt      2026-08-04 12:09:42.767553595 +0000  
+++ /tmp/puppet-file20260804-2278-va5q31 2026-08-04 12:18:59.313048615 +0  
000  
@@ -14,7 +14,7 @@  
  OS Family: Debian  
  
  Kernel: Linux  
-Architecture:  
+Architecture: amd64  
  
  Processors: 2  
  Memory: 914.01 MiB  
  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/con
```

```
ubuntu@puppet-agent01: ~  
  
Processors: 2  
Memory: 914.01 MiB  
  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/con  
tent: content changed '{sha256}0032a8b40c698e91ce2d91f734adad1774a47617c8841c660  
2930a81fed2121f' to '{sha256}3cae9473383acedf08ff0eea520a378e06c14cab7e28283785c  
e0605e73f8e73'  
[Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]: Sche  
duling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh'  
from 1 event  
Notice: Applied catalog in 0.57 seconds  
ubuntu@puppet-agent01:~$ cat /opt/company/system-info.txt  
-----  
Enterprise Infrastructure Report  
-----  
  
Server Role: Web Server  
Environment: Production  
  
Hostname: puppet-agent01  
FQDN: puppet-agent01.ap-south-1.compute.internal  
Primary IP: 172.31.7.84
```

```
ubuntu@puppet-agent01: ~  
  
Environment: Production  
  
Hostname: puppet-agent01  
FQDN: puppet-agent01.ap-south-1.compute.internal  
Primary IP: 172.31.7.84  
  
Operating System: Ubuntu  
OS Release: 22.04  
OS Family: Debian  
  
Kernel: Linux  
Architecture: amd64  
  
Processors: 2  
Memory: 914.01 MiB  
  
Puppet Version: 8.10.0  
  
Generated By: Puppet Master and Vishnu's Passion  
  
-----  
ubuntu@puppet-agent01:~$ |
```

Agent-02:

```
ubuntu@puppet-agent02: ~  
Last login: Tue Aug  4 07:38:46 2026 from 128.185.168.204  
ubuntu@puppet-agent02:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent02  
Info: Applying configuration version '1785845449'  
Notice: Applied catalog in 0.48 seconds  
ubuntu@puppet-agent02:~$ ls -l /cat/company  
ls: cannot access '/cat/company': No such file or directory  
ubuntu@puppet-agent02:~$ ls -l /opt/company  
total 8  
-rw-r--r-- 1 root root  68 Aug  4 12:09 os-info.txt  
-rw-r--r-- 1 root root 554 Aug  4 12:09 system-info.txt  
ubuntu@puppet-agent02:~$ cat /opt/os-info.txt  
cat: /opt/os-info.txt: No such file or directory  
ubuntu@puppet-agent02:~$ cat /opt/company/os-info.txt  
This node belongs to the Debian family.  
Package Manager: APT  
ubuntu@puppet-agent02:~$ cat /opt/company/system-info.txt  
-----
```

```
ubuntu@puppet-agent02: ~  
ubuntu@puppet-agent02:~$ cat /opt/company/os-info.txt  
This node belongs to the Debian family.  
Package Manager: APT  
ubuntu@puppet-agent02:~$ cat /opt/company/system-info.txt  
-----  
Enterprise Infrastructure Report  
-----  
  
Server Role: Application server  
Environment: Development  
  
Hostname: puppet-agent02  
FQDN: puppet-agent02.ap-south-1.compute.internal  
Primary IP: 172.31.11.239  
  
Operating System: Ubuntu  
OS Release: 22.04  
OS Family: Debian  
  
Kernel: Linux  
Architecture:  
  
Processors: 2  
Memory: 914.01 MiB
```

```
ubuntu@puppet-agent02: ~  
Memory: 914.01 MiB  
Puppet Version: 8.10.0  
Generated By: Puppet Master and Vishnu's Passion  
  
-----  
ubuntu@puppet-agent02:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent02  
Info: Applying configuration version '1785845993'  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/con  
tent:  
--- /opt/company/system-info.txt          2026-08-04 12:09:44.581402892 +0000  
+++ /tmp/puppet-file20260804-2097-ftqae  2026-08-04 12:19:54.597404427 +0  
000  
@@ -14,7 +14,7 @@  
OS Family: Debian
```

```
ubuntu@puppet-agent02: ~  
Kernel: Linux  
-Architecture:  
+Architecture: amd64  
  
Processors: 2  
Memory: 914.01 MiB  
  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/con  
tent: content changed '{sha256}3fb15f024b7d8e5d485cb3f81810e03acd93961ec0db6574e  
cfff2896f40bd5c9' to '{sha256}72680b4eb044372c0eb4a38f20144b0297baeefacf7eb9112e6  
c0ef599fcfc4e'  
Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]: Sche  
duling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh'  
from 1 event  
Notice: Applied catalog in 0.64 seconds  
ubuntu@puppet-agent02:~$ cat /opt/company/system-info.txt  
-----  
Enterprise Infrastructure Report  
-----  
  
Server Role: Application server  
Environment: Development
```

```
ubuntu@puppet-agent02: ~  
FQDN: puppet-agent02.ap-south-1.compute.internal  
Primary IP: 172.31.11.239  
  
Operating System: Ubuntu  
OS Release: 22.04  
OS Family: Debian  
  
Kernel: Linux  
Architecture: amd64  
  
Processors: 2  
Memory: 914.01 MiB  
  
Puppet Version: 8.10.0  
Generated By: Puppet Master and Vishnu's Passion  
  
-----  
ubuntu@puppet-agent02:~$ cat /opt/company/os-info.txt  
This node belongs to the Debian family.  
Package Manager: APT  
ubuntu@puppet-agent02:~$
```

